

Empiric Use of Ampicillin and Cefotaxime, Compared With Ampicillin and Gentamicin, for Neonates at Risk for Sepsis Is Associated With an Increased Risk of Neonatal Death

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ABSTRACT

BACKGROUND. We reported previously that the use of cephalosporin among premature neonates increased the risk of subsequent fungal sepsis. As a result, we recommended that ampicillin and gentamicin be used as empiric coverage for early-onset neonatal sepsis while culture results are awaited.

OBJECTIVES. To describe antibiotic use during the first 3 days after birth for neonates admitted to the NICU and to evaluate the outcomes for neonates treated with 2 different antibiotic regimens.

METHODS. We assembled a cohort of inborn neonates, from our deidentified administrative database, who had documented exposure to ampicillin during the first 3 days after birth. Infants treated concurrently with cefotaxime or gentamicin were evaluated, to identify the factors that were associated independently with death before discharge, with both univariate and multivariate analyses.

RESULTS. There were 128 914 neonates selected as the study cohort; 24 111 were treated concurrently with ampicillin and cefotaxime and 104 803 were treated concurrently with ampicillin and gentamicin. Logistic modeling showed that neonates treated with ampicillin/cefotaxime were more likely to die (adjusted odds ratio: 1.5; 95% confidence interval: 1.4–1.7) and were less likely to be discharged to home or foster care than were neonates treated with ampicillin/gentamicin. This observation was true across all estimated gestational ages. Other factors that were associated independently with death included immature gestational age, need for assisted ventilation on the day of admission to the NICU, indications of perinatal asphyxia or major congenital anomaly, and reported use of ampicillin/cefotaxime.

CONCLUSIONS. For patients receiving ampicillin, the concurrent use of cefotaxime during the first 3 days after birth either is a surrogate for an unrecognized factor or is itself associated with an increased risk of death, compared with the concurrent use of gentamicin.

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Key Words

antibiotics, neonate, mortality, sepsis

Abbreviations

OR—odds ratio

CI—confidence interval

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CULTURE-PROVEN NEONATAL SEPSIS is a serious but uncommon problem. The risk of delays in diagnosis or treatment leads to admissions to NICUs for sepsis evaluations and empiric antibiotic treatment.^{1,2} Because culture-proven neonatal sepsis is associated with increased mortality rates, morbidity, and prolonged hospital stays, both the human and fiscal costs of this disease are high.² Decisions about how to prevent neonatal sepsis, who and how long to treat, and which antibiotics to use remain important clinical problems.³⁻⁶

The choice of antibiotic treatment for neonatal sepsis must be driven by hospital-specific guidelines based on prevalent organisms and their susceptibility patterns in the particular nursery/hospital environment. The choice of perinatal and early-onset neonatal antimicrobial agents may facilitate the appearance of organisms that cause late-onset neonatal sepsis and change the antibiotic resistance patterns of organisms that cause early-onset sepsis.^{4,7,8} Previously we reported that cephalosporin or carbapenem use for premature neonates may increase the risk of subsequent fungal sepsis.⁹ Therefore, we recommended that ampicillin and gentamicin be used for empiric treatment of early-onset neonatal sepsis while culture results are awaited.¹⁰

As part of ongoing development of a data warehouse by Pediatrix Medical Group (Sunrise, FL), we have structured the storage of medication data for retrieval and analysis. In our first evaluation of medication use and variation, we noted that antibiotics are the most commonly used medications in the NICU. We decided to confirm the impact of our previous empiric antibiotic recommendations for suspected early-onset sepsis and noted that a significant proportion (>10%) of neonates who were cared for in our NICUs were still being treated with a combination of ampicillin and cefotaxime. The purpose of this study was to describe the outcomes of 2 different antibiotic regimens for neonates treated in the NICU and to determine which factors were associated most closely with those outcomes.

METHODS

Study Cohort

We assembled a retrospective cohort of neonates from the Pediatrix deidentified administrative data set, from which several other studies have been published,¹¹⁻¹⁴ to compare patients treated concurrently with ampicillin and cefotaxime with patients treated concurrently with ampicillin and gentamicin. Patients were included in the cohort only if they were inborn (to reduce the potential confounding effect of transport on outcomes) and had antibiotic treatment initiated within 3 days after birth.

Clinical data on these neonates were recorded during the time that care was being provided in the NICU. Admission, discharge, and daily progress notes are generated with a computer-assisted tool, and the data are

stored in an electronic database. These data are consolidated within the Pediatrix data warehouse, where they are deidentified, made compliant with the Health Insurance Portability and Accountability Act of 1996, and configured into tables that can be joined and queried for statistical analyses. Specific tables used for this analysis were "patients," "admissions," "medications," "diagnoses," and "cultures." The use of the data reported here and certification of the data as deidentified were approved by the institutional review board of the Wichita Medical Research and Education Foundation (Wichita, KS).

Data on estimated gestational ages represented the best estimates based on both obstetric data and neonatal examination findings. Data on race were based on the options contained in the database, ie, white, black, Hispanic, Native American, and Asian. Other data elements used in this analysis are listed in Table 1. We evaluated diagnostic patterns (diagnoses made in the first 21 days after birth) and types of bacteria reported from positive cultures (from blood or cerebrospinal fluid) documented during the hospitalization and the first 7 days after birth (Tables 1 and 2), in an attempt to identify a pattern of potential adverse events that would explain the increased risk of death. Each diagnosis or culture was counted only once per patient, but a patient with >1 diagnosis or culture could have been counted more than once. Data were normalized to the total number of patients in each specific antibiotic group and not to the total number of diagnoses or reports. Reports of diagnoses and cultures reflect the total number of patients within a specific antibiotic treatment group who had a report of a specific diagnosis or culture type. A single patient could be counted more than once, and the percentages (total number of occurrences of a specific diagnosis or culture report in that antibiotic group) represent the occurrence rates of the diagnosis or culture report in each group. As a result, the percentages in these categories add up to >100%. The opposite is true of outcome and survival data; each patient was counted only once.

We also counted any report of a persistent hearing deficit or abnormal auditory evoked brainstem response as a "hearing problem." These results are noted in Table 1.

Survey

Using the Pediatrix internal E-mail system, we posted a survey with the following questions concerning the use of cefotaxime for neonates with suspected sepsis. (1) Do you prescribe cefotaxime within the first 3 days after birth? (2) Do you use ampicillin/cefotaxime or ampicillin/gentamicin as your first line of treatment for neonates with suspected sepsis (routine or option)? (3) Estimate the proportion of neonatal admissions treated with ampicillin/gentamicin. (4) Describe situations in which you would pick ampicillin/cefotaxime over ampi-

TABLE 1 Population Characteristics and Outcomes

Variable	Ampicillin/Cefotaxime	Ampicillin/Gentamicin
No.	24 111	104 803
Maternal age ^a		
Missing data, <i>n</i> (%)	4 (0.02)	46 (0.04)
Median (25th–75th percentile), y	28 (23–32)	27 (24–33)
Estimated gestational age ^a		
Missing data, <i>n</i> (%)	2 (0.01)	7 (0.01)
Median (25th–75th percentile), wk	35 (31–38)	35 (32–38)
Birth weight ^a		
Missing data, <i>n</i> (%)	61 (0.25)	220 (0.21)
Median (25th–75th percentile), kg	2.35 (1.5–3.2)	2.42 (1.7–3.2)
Apgar score at 1 min		
Missing data, <i>n</i> (%)	116 (0.48)	625 (0.6)
Median (25th–75th percentile)	7 (5–8)	8 (6–8)
Apgar score at 5 min		
Missing data, <i>n</i> (%)	118 (0.49)	633 (0.6)
Median (25th–75th percentile)	9 (8–9)	9 (8–9)
Day antibiotics started		
Missing data, <i>n</i> (%)	0	0
Mean $\pm$ SD	0.2 $\pm$ 0.5	0.2 $\pm$ 0.5
Fractional inspired oxygen on day of admission		
Missing data, <i>n</i> (%)	1547 (6)	5328 (5)
Median (25th–75th percentile)	0.3 (0.2–0.6)	0.3 (0.2–0.5)
Duration of antibiotic therapy		
Missing data, <i>n</i> (%)	1990 (8)	5962 (6)
Median (25th–75th percentile), d	3 (2–7)	3 (2–6)
Race, <i>n</i> (%)		
Missing data	908 (3.8)	3806 (3.6)
Black	3920 (16.3)	16 455 (15.7)
Hispanic	3856 (16)	25 474 (24.3)
Other	1410 (5.8)	5270 (5.0)
White	14 017 (58.1)	53 798 (51.3)
Delivered through cesarean section, <i>n</i> (%)	11 799 (48.9)	50 991 (48.7)
Report of positive test for group B <i>Streptococcus</i> for patient's mother, <i>n</i> (%)	3530 (14.6)	13 958 (13.3)
Any report of positive blood or cerebrospinal fluid culture during hospitalization, <i>n</i> (%)	1544 (6.4)	6296 (6)
Positive blood or cerebrospinal fluid culture in first 7 d, <i>n</i> (%)	542 (2.2)	2465 (2.4)
Most common bacteria associated with report of positive blood or cerebrospinal fluid culture (first 7 d), <i>n</i> (%) ^b		
Coagulase-negative <i>Staphylococcus</i>	139 (0.58)	827 (0.79)
Group B <i>Streptococcus</i>	82 (0.34)	388 (0.37)
Gram-positive cocci	62 (0.26)	280 (0.27)
<i>Escherichia coli</i>	26 (0.11)	128 (0.12)
<i>Staphylococcus</i> species	15 (0.06)	69 (0.07)
<i>Streptococcus viridians</i>	22 (0.09)	66 (0.06)
<i>Staphylococcus aureus</i>	12 (0.05)	55 (0.05)
<i>Enterococcus</i>	8 (0.03)	48 (0.05)
Gram-negative rod	7 (0.03)	40 (0.04)
Degree of respiratory support on day of admission, <i>n</i> (%) ^a		
Missing data	334 (1.4)	1252 (1.2)
Continuous positive airway pressure	2611 (10.8)	14 483 (13.8)
High-frequency ventilation	1729 (7.2)	4369 (4.2)
Hood oxygen	4207 (17.4)	21 711 (20.7)
Nasal cannula	1415 (5.9)	6147 (5.9)
Room air	7409 (30.7)	34 915 (33.3)
Ventilator	6406 (26.6)	21 926 (20.9)
Report of depression, <i>n</i> (%) ^a	2169 (9)	4936 (4.7)
Depression type, <i>n</i> (%) ^b		
Asphyxia	81 (0.3)	131 (0.1)
Birth asphyxia	111 (0.5)	157 (0.1)
Depression at birth	440 (1.8)	866 (0.8)
Hypoxic-ischemic encephalopathy	206 (0.9)	224 (0.2)
Neonatal depression	160 (0.7)	334 (0.3)
Perinatal asphyxia	104 (0.4)	145 (0.1)
Perinatal depression	851 (3.5)	2061 (2)
Seizures occurring during the 7 d after birth	633 (2.6)	1514 (1.4)

TABLE 1 Continued

Variable	Ampicillin/Cefotaxime	Ampicillin/Gentamicin
Report of anomaly, <i>n</i> (%)	2257 (9.4)	8766 (8.4)
Anomalies reported most often, <i>n</i> (%)		
Trisomy 21	164 (0.7)	722 (0.7)
Gastroschisis	122 (0.5)	702 (0.7)
Heart disease	54 (0.2)	314 (0.3)
Cardiomyopathy	80 (0.3)	274 (0.3)
Myelomeningocele	68 (0.3)	279 (0.3)
Supraventricular tachycardia	53 (0.2)	250 (0.2)
Diaphragmatic hernia	53 (0.2)	227 (0.2)
Chromosomal abnormality	60 (0.2)	204 (0.2)
Congenital adrenal hyperplasia	29 (0.1)	231 (0.2)
Diagnosis (before 21 d of age; top 10 reported), <i>n</i> (%) ^b		
Sepsis ruled out	9784 (40.6)	47 820 (45.6)
Sepsis possible	10 804 (44.8)	40 380 (38.5)
Respiratory distress syndrome	6951 (28.8)	27 740 (26.5)
Respiratory distress	6275 (26)	27 823 (26.5)
Sepsis	2868 (11.9)	17 562 (16.8)
Hypoglycemia	2765 (11.5)	13 788 (13.2)
Transient tachypnea	1422 (5.9)	10 486 (10)
Hypotension	2995 (12.4)	10 238 (9.8)
Thrombocytopenia	1931 (8)	6853 (6.5)
Prematurity, extreme	1731 (7.2)	5916 (5.6)
Status, <i>n</i> (%) ^a		
Transfer	2796 (11.6)	11 250 (10.7)
Alive (≥ 28 d after birth)	20 304 (84.2)	91 510 (87.3)
Died (≤ 28 d after birth)	1011 (4.2)	2043 (1.9)
Status at discharge, <i>n</i> (%) ^a		
Missing data	33 (0.13)	88 (0.08)
Died	1140 (4.7)	2433 (2.3)
Home or foster care	19 603 (81.3)	89 271 (85.2)
Transfer	3335 (13.9)	13 011 (12.4)
Report of hearing problem before discharge, <i>n</i> (%)	236 (0.98)	1195 (1.14)
Age at discharge		
Missing data, <i>n</i> (%)	0	0
Median (25th–75th percentile), d	10 (5–26)	10 (5–23)

^a $P < .01$.^b A diagnosis or culture was counted only once per patient, but a patient with >1 diagnosis or culture could be counted more than once. Data were normalized to the total number of patients and not to the total number of reports.

illin/gentamicin as your first line of treatment. The goal of this informal survey was to determine what factors might alter outcomes and to identify patient selection bias for the patients treated with ampicillin/cefotaxime.

Analyses

Univariate Analyses

We compared the 2 study populations by using both univariate and multivariate techniques. Continuous variables (estimated gestational age and birth weight) were evaluated with 2-tailed *t* tests. Categorical variables (eg, race and gender) were evaluated with 2-tailed χ^2 tests. Nonparametric continuous data were assessed with Kruskal-Wallis analysis of variance.

After making the initial observation that mortality risk seemed unequal, we evaluated potential selection bias immediately. Using the Pediatrix internal E-mail system, we circulated a survey to clinicians affiliated with Pediatrix, to try to determine what factors influ-

enced their use of ampicillin/cefotaxime rather than ampicillin/gentamicin. The main reasons indicated for the choice of cefotaxime over gentamicin as initial therapy for sepsis were the potential nephrotoxic or ototoxic effects of gentamicin, selective use of cefotaxime for neonates with a history of neonatal depression and/or perinatal asphyxia, and suspected or culture-proven Gram-negative organisms causing sepsis or meningitis. These factors and those found in univariate analyses to differ between the 2 antibiotic groups were used to develop a logistic model that identified risk factors associated independently with death.

Multivariate Analyses

After univariate analyses, we used multivariate logistic regression to identify factors associated independently with death before discharge. In the logistic regression analysis, we incorporated the variables (Table 1) found in univariate analyses to be significantly different for the

TABLE 2 Diagnoses Made in First 21 Days for Which the Rate Reported for the Ampicillin/Cefotaxime Group Was ≥ 1.5 or ≤ 0.5 Times That Reported for the Ampicillin/Gentamicin Group

Variable	n (%)		Relative Rate
	Ampicillin/Cefotaxime	Ampicillin/Gentamicin	
Liver dysfunction	96 (0.398)	158 (0.151)	2.6
Renal failure	237 (0.983)	394 (0.376)	2.6
Cardiac arrest	154 (0.639)	287 (0.274)	2.3
Coagulopathy	310 (1.286)	609 (0.581)	2.2
Disseminated intravascular coagulation	131 (0.543)	269 (0.257)	2.1
Renal dysfunction	189 (0.784)	399 (0.381)	2.1
Alkalosis	35 (0.145)	81 (0.077)	1.9
Hydrocephalus (posthemorrhagic)	118 (0.489)	271 (0.259)	1.9
Pulmonary hemorrhage	344 (1.427)	800 (0.763)	1.9
Perinatal depression	848 (3.517)	2061 (1.967)	1.8
Intraventricular hemorrhage (grade IV)	257 (1.066)	673 (0.642)	1.7
Perforated bowel	85 (0.353)	218 (0.208)	1.7
Tachycardia	107 (0.444)	270 (0.258)	1.7
Hypertension	269 (1.116)	718 (0.685)	1.6
Hypovolemia	616 (2.555)	1669 (1.593)	1.6
Intracranial hemorrhage	129 (0.535)	357 (0.341)	1.6
Intraventricular hemorrhage (grade III)	311 (1.29)	831 (0.793)	1.6
Patent ductus arteriosus (ligation)	352 (1.46)	957 (0.913)	1.6
Seizures (during first 21 d)	737 (3.057)	1947 (1.858)	1.6
Congestive heart failure	45 (0.187)	128 (0.122)	1.5
Hyperkalemia	423 (1.754)	1221 (1.165)	1.5
Necrotizing enterocolitis (surgical)	82 (0.34)	280 (0.267)	1.3
Necrotizing enterocolitis (medical)	97 (0.402)	867 (0.827)	0.5

2 groups ($P < .1$). Variables were entered into the model with a stepwise selection method (for entry and retention, $P < .1$). Maternal age, birth weight, and gestational age were entered into the model as continuous variables. The need for assisted ventilation on the day of admission to the NICU, a report suggesting perinatal asphyxia (see list in Table 1) or a report of a major anomaly (see partial list of most common anomalies in Table 1), and the reported use of ampicillin/cefotaxime were entered into the model as dichotomous ("yes/no" or "present/absent") categorical variables. The site of care (the hospital where the patient was treated) was evaluated as a categorical variable in the logistic model. In addition, we

defined 3 types of sites of care, ie, low-use sites, where ampicillin/cefotaxime was used for $\leq 30\%$ of the patients cared for; moderate-use sites, where ampicillin/cefotaxime was used for 31% to 59% of the patients cared for; and high-use sites, where ampicillin/cefotaxime was used for $\geq 60\%$ of the patients cared for. Cases with missing values for any of the independent variables were excluded from the analyses. The number of patients evaluated in the logistic model is listed with the results in Table 3.

RESULTS

Between January 1, 1996, and June 1, 2004, care was provided by clinicians affiliated with Pediatrix Medical Group to 227 711 neonates, of whom 192 989 (85%) were born at the site where neonatal intensive care was provided. The other 34 722 neonates (15%) were transported for care or their care represented a subsequent admission.

We identified 1 338 980 reports of 409 different medications during the time period from January 1996 through June 2004. For these data, an antibiotic course could be represented as a single report with start and stop dates or as sequential reports that represented each day of antibiotic administration. There were a total of 172 930 reports of ampicillin use, with 161 049 (93.1%) reported in the first 3 days after birth. There were 156 384 reports of gentamicin use, 130 616 (83.5%) of which occurred in the first 3 days after birth; 54 255 reports of cefotaxime use, 34 168 (63.0%) of which occurred in the first 3 days after birth; and 9965 reports of tobramycin use, 2716 (27%) of which occurred in the first 3 days after birth. Reports of these 4 antibiotics were made for a total of 159 636 unique patients derived from the medications table, which included outborn patients. We then matched these reports to our inborn demographic data set.

For the inborn neonates ($n = 192\,989$) and the patients with reported use of 1 of the 4 antibiotics (ampicillin, cefotaxime, gentamicin, and tobramycin) in the first 3 days after birth ($n = 159\,636$), there were 135 095 matches. Of these matches, 128 914 neonates were se-

TABLE 3 Adjusted ORs (From Logistic Model)

Variable	OR (95% CI)	
	All Patients in the Model ^a	Only Neonates With EGA of >37 wk and Birth Weight of >2 kg ^b
No. of patients in the model	111 000	40 704 ^c
Treated with ampicillin/cefotaxime	1.5 (1.4–1.7)	1.5 (1.1–2)
Report of assisted ventilation on admission day	9.4 (8.1–10.9)	20.5 (15.2–28.1)
Report of history of depression	4.9 (4.3–5.5)	4 (3.1–5.3)
Report of major anomaly	5.1 (4.6–5.6)	12 (9.2–15.7)

EGA indicates estimated gestational age.

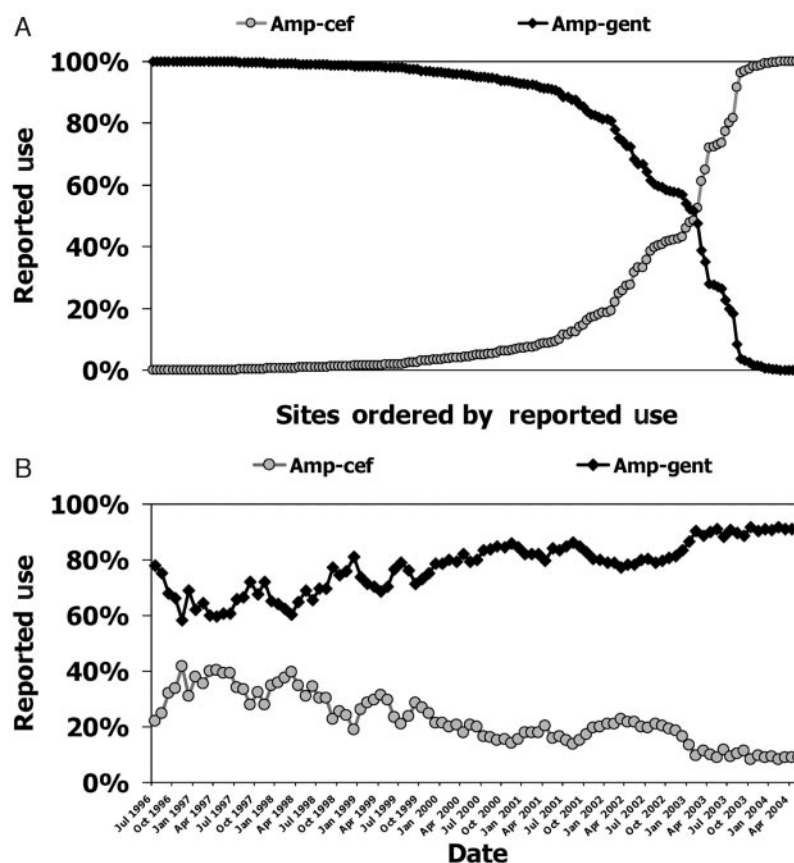
^a Estimated gestational age in the model as a continuous variable.

^b Birth weight in the model as a continuous variable.

^c Excludes neonates whose discharge was to another hospital (transfers).

FIGURE 1

Reported use of ampicillin/cefotaxime (Amp-cef), compared with ampicillin/gentamicin (Amp-gent), according to NICU site of care (A) and change in relative use over time (B).



lected for the study cohort as a result of their antibiotic use; 24 111 were treated concurrently with ampicillin and cefotaxime, and 104 803 were treated concurrently with ampicillin and gentamicin. These 2 strategies were the most commonly reported strategies for the empiric treatment of neonates considered at risk for neonatal sepsis.

There was significant site variation in the reported use of ampicillin/cefotaxime and ampicillin/gentamicin (Fig 1A). Twenty-five (15%) of the 165 sites that reported data on >100 patients used ampicillin/cefotaxime for >50% of their patients. In addition to site variation, there were differences in the relative use of these 2 treatment approaches over time, with a decrease in the overall use of ampicillin/cefotaxime between 1996 and 2004 (Fig 1B).

A survey was sent via an internal E-mail system, which we estimated reached ~200 clinicians, on the basis of review of the message history. One hundred twenty-one clinicians affiliated with Pediatrix responded to this survey, which was circulated to determine the factors that influenced their use of ampicillin/cefotaxime rather than ampicillin/gentamicin. Forty-six (38%) reported some use of cefotaxime for treatment of neonates and 4 reported routine use of cefotaxime for treatment of neonates admitted to their NICU. The main reasons indicated for the choice of cefotaxime over gentamicin as

initial therapy for sepsis were the potential nephrotoxic or ototoxic effects of gentamicin, selective use of cefotaxime for neonates with a history of neonatal depression and/or perinatal asphyxia, and suspected or culture-proven Gram-negative organisms causing sepsis or meningitis. Two clinicians raised concerns about the potential interaction between gentamicin and magnesium, causing hypotonia and respiratory depression.

Univariate analyses showed that neonates treated with ampicillin/cefotaxime were slightly more immature, had lower birth weights, were reported more often to be white and less often to be Hispanic, were reported more often to require mechanical ventilation on the day of admission, and were reported more often to have perinatal and/or neonatal depression (interestingly, there were no significant differences in Apgar scores), compared with neonates who were treated with ampicillin/gentamicin (Table 1). Diagnoses made before 21 days of age were examined in an attempt to identify a pattern of potential adverse events. There was no difference in the patterns of the most common (top 10) diagnoses (Table 1). In addition, reports of major anomalies and positive blood and/or cerebrospinal fluid cultures were similar. We did find a group of relatively rare (<5% of the total sample) diagnoses that occurred at different rates (the rate reported for the ampicillin/cefotaxime group was ≥ 1.5 or ≤ 0.5 times that reported for

the ampicillin/gentamicin group) for the 2 treatment groups (Table 2). We have provided these descriptive data for informational purposes only; we did not attempt to correct for the effects of gestational age and birth weight on the relative rates of occurrence of these events.

Neonates treated with ampicillin/cefotaxime were more likely to die and were less likely to be discharged to home or foster care than were neonates treated with ampicillin/gentamicin (Table 1). These findings were true across all estimated gestational ages. Factors that were found to be associated independently with death included an immature gestational age, the need for assisted ventilation on the day of admission to the NICU, a report suggesting perinatal asphyxia or a report of a major anomaly, and the reported use of ampicillin/cefotaxime (Table 3). Entering site of care into the model as either a categorical variable or a group variable (low-, moderate-, or high-use sites, as defined above) did not alter these results.

When we limited regression model development to neonates with an estimated gestational age of ≥ 37 weeks and a birth weight of >2 kg, the same variables were found to be important, except that birth weight, rather than gestational age, was retained in the logistic regression model. The calculated odds ratios (ORs) were similar in the 2 analyses (Table 3). To evaluate the ORs within specific gestational-age groups, the final model was analyzed within gestational-age groups; these results are reported graphically (Fig 2).

DISCUSSION

We found that antibiotics were the most common medications reported in the NICU and that most antibiotic use was initiated during the first 3 days after birth. In our study cohort, only 2% had a report of a positive culture during the first 7 days after birth and $<1\%$ had a report of group B *Streptococcus*, *Escherichia coli*, or other known serious neonatal pathogens. Most patients (98%) did not have culture-proven sepsis.

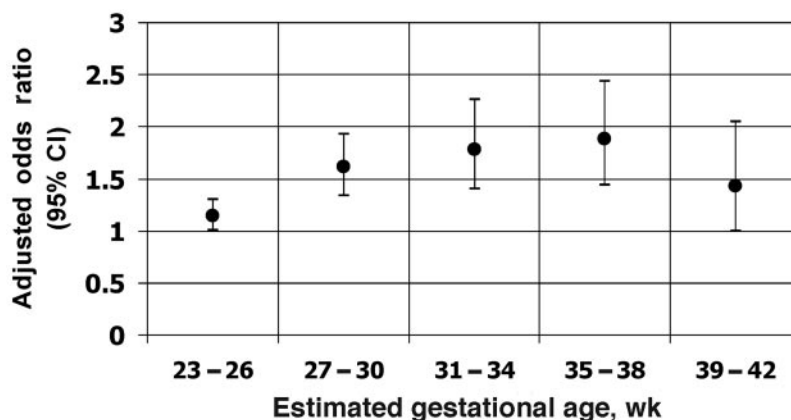
We anticipated that most of the antibiotics administered during the first 3 days after birth would be prescribed for neonates who were identified as being "at risk for sepsis" and that mortality rates for these neonates would be low and similar, independent of the empiric antibiotics used. We found that most antibiotic use was for neonates at risk for sepsis and that overall mortality rates were low. Consequently, it was surprising to find that mortality rates for ampicillin-treated neonates were higher when ampicillin was combined with cefotaxime, compared with gentamicin.

After making the initial observation that mortality risk seemed unequal, we evaluated potential selection bias immediately. Although the majority of sites used predominantly ampicillin and gentamicin concurrently, 15% of the 165 sites that reported data for >100 patients administered ampicillin and cefotaxime concurrently for $>50\%$ of their patients. On the basis of these data, we circulated a survey to clinicians affiliated with Pediatrix, to determine their primary reasons for choosing cefotaxime over gentamicin. This survey demonstrated 2 potential confounding factors driven by selection bias. Clinicians reported selective use of ampicillin/cefotaxime for neonates who had a history suggesting Gram-negative sepsis or a report of perinatal/neonatal depression/asphyxia.

These findings led us to evaluate the impact of several important confounding variables on our observation of increased mortality rates. These variables included all of the items listed in Table 1. We expected that logistic regression analysis that included these variables would allow us to exclude antibiotic choice (ampicillin/cefotaxime versus ampicillin/gentamicin) as an associated risk factor. Our results, however, did not permit this exclusion (Table 3 and Fig 2). Consequently, we are compelled to report that the use of ampicillin/cefotaxime (in the NICU) during the first 3 days after birth may be associated with an increased risk of death (adjusted OR: 1.5; 95% confidence interval [CI]: 1.4–1.7), compared with the use of ampicillin/gentamicin.

FIGURE 2

Adjusted OR (based on final model) within gestational-age groups (logistic regression [odds of death] adjusted for need for assisted ventilation, anomalies, birth depression, and estimated gestational age [EGA] within each estimated gestational-age group).



We realize the limitations of making this observation on the basis of retrospective review of an administrative data set. Proxies for selection bias may not reflect adequately the true severity of illness or the therapeutic approach. There may be variations in the process of care other than the specific choice of antibiotics that influence the differences in mortality rates we report. Retrospective studies are also limited by incomplete data, and the evaluation of interactions between drugs is better performed prospectively. However, validation of the observations reported here would require an exceptionally large clinical trial, which is unlikely to be performed. Therefore, clinical practice for this group of newborns at risk for neonatal sepsis will likely be driven by observational data similar to the findings we report here. These data have been distributed to clinicians affiliated with Pediatrix, and we are continuing to monitor actively both the use of cefotaxime and the reported mortality rates for neonates treated with these 2 combinations of antibiotics.

CONCLUSIONS

Our data demonstrate that a commonly used empiric antibiotic regimen may be associated with an increased risk of death. This report is confounded by the observation that the most important risk factors associated with death are premature birth, low birth weight, the need for assisted ventilation on the day of admission, the presence of major anomalies, and a history suggesting significant perinatal hypoxia. The use of ampicillin/cefotaxime may be driven, in part, by these factors, resulting in selection bias. However, our logistic modeling of these factors did not eliminate the possibility that, among neonates receiving ampicillin in the first 3 days after birth, the concurrent use of cefotaxime (or something for which this is a surrogate measure) increases the risk of death. We suggest that neonatal health care professionals examine their own practices carefully and encourage others strongly to validate or to dismiss our observations.

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